

1.8 Introduction to Linear Transformations

I

A transformation (or function or mapping) T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a rule that assigns to each vector $\vec{x}$ in $\mathbb{R}^n$ a vector $T(\vec{x})$ in $\mathbb{R}^m$. The set $\mathbb{R}^n$ is called the domain of T , and $\mathbb{R}^m$ is called the codomain of T . The notation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ indicates that the domain of T is $\mathbb{R}^n$ and the codomain is $\mathbb{R}^m$. For $\vec{x}$ in $\mathbb{R}^n$, the vector $T(\vec{x})$ in $\mathbb{R}^m$ is called the image of $\vec{x}$ (under the action of T). The set of all images $T(\vec{x})$ is called the range of T .

Let's try to drive this home with an example.

$$\text{Let } A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}, \quad \vec{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad \vec{b} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}, \quad \vec{c} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}, \text{ and}$$

$(m \times n) \quad m \times n \quad m \times n \quad m \times n$
 $(3 \times 2) \quad (2 \times 1) \quad (3 \times 1) \quad (3 \times 1)$

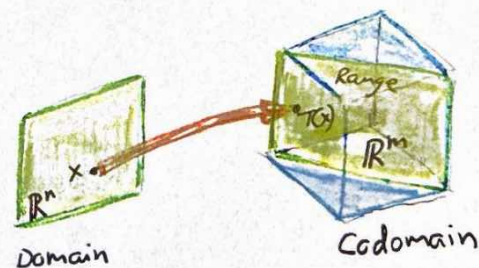
define a transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by $T(\vec{x}) = A\vec{x}$, so that

$$T(\vec{x}) = A\vec{x} = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1x_1 - 3x_2 \\ 3x_1 + 5x_2 \\ -1x_1 + 7x_2 \end{bmatrix}$$

a. Find $T(\vec{u})$, the image of $\vec{u}$ under the transformation T .

To find $T(\vec{u})$ we compute $A\vec{u}$

$$T(\vec{u}) = A\vec{u} = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 + 3 \\ 6 - 5 \\ -2 - 7 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ -9 \end{bmatrix}$$



Domain, codomain and range of $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$

For simplicity we denote matrix transformations as $\vec{x} \mapsto A\vec{x}$.

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II

b. Find an $\vec{x}$ in $\mathbb{R}^2$ whose image under T is $\vec{b}$.

They want us to solve $T(\vec{x}) = \vec{b}$ for $\vec{x}$. Since we have defined $T(\vec{x}) = A\vec{x}$, we will solve the equation

$$\begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$$

for $\vec{x}$ using elimination by matrix.
Setting up the augmented matrix and performing row reduction.

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & -5 \end{array} \right]$$

$$R_2 \leftarrow R_2 - 3R_1$$

$$R_3 \leftarrow R_3 + R_1$$

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & -2 \end{array} \right]$$

$$R_2 \leftarrow R_2 \cdot 1/14$$

$$R_3 \leftarrow R_3 \cdot 1/4$$

$$R_3 \leftarrow R_3 - R_2$$

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 1 & -1/2 \\ 0 & 0 & 0 \end{array} \right]$$

$$R_1 \leftarrow R_1 + 3R_2$$

$$\left[\begin{array}{cc|c} 1 & 0 & 3/2 \\ 0 & 1 & -1/2 \\ 0 & 0 & 0 \end{array} \right]$$

Row reduction gives

$$x_1 = 3/2$$

$$x_2 = -1/2$$

$$\boxed{\vec{x} = \begin{bmatrix} 3/2 \\ -1/2 \end{bmatrix}}$$

Testing result on A

$$\begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 3/2 \\ -1/2 \end{bmatrix} = \begin{bmatrix} 3/2 + 3/2 \\ 9/2 - 5/2 \\ -3/2 - 7/2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$$

This produces $\vec{b}$, which indicates the solution is correct.

We also take note of that we have no free variable, such that our solution

$$\vec{x} = \begin{bmatrix} 3/2 \\ -1/2 \end{bmatrix}$$

is the only solution to the equation

$$T(\vec{x}) = \vec{b}$$

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III

c. Is there more than one $\vec{x}$ whose image under T is $\vec{b}$? No, the solution we found in 5. is unique.

d. Determine if $\vec{c}$ is in the range of the transformation T .

What they're asking is if $A\vec{x} = \vec{c}$ is consistent, if it has a solution. If $\vec{c}$ is an image of $\vec{x}$ under the action of T . Let's set up an augmented matrix and check.

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & 5 \end{array} \right]$$

$$R_2 \leftarrow R_2 - 3R_1$$

$$R_3 \leftarrow R_3 + R_1$$

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & 8 \end{array} \right]$$

$$R_2 \leftarrow R_2 \cdot 1/14$$

$$R_3 \leftarrow R_3 \cdot 1/4$$

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 1 & -1/2 \\ 0 & 1 & 2 \end{array} \right]$$

$$R_3 \leftarrow R_3 - R_2$$

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 1 & -1/2 \\ 0 & 0 & 5/2 \end{array} \right]$$

Row reduction give us a last line that's impossible, $0 = 5/2$ is not valid.

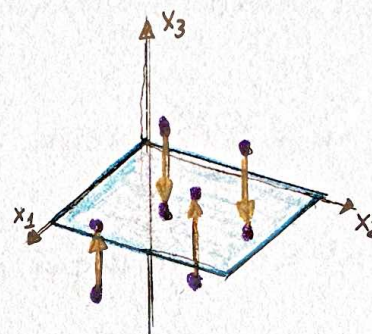
Thus, the system is inconsistent.

$\vec{c}$ is not in the range of T .

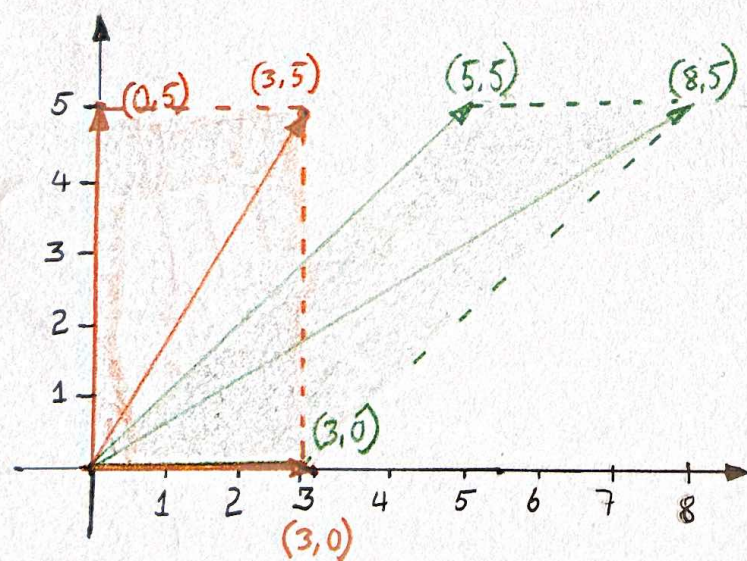
Example 2

If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, then the transformation $\vec{x} \mapsto A\vec{x}$ projects points in $\mathbb{R}^3$ onto the x_1x_2 -plane because

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ 0 \end{bmatrix}$$



Example 3



Let $A = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$. The transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$T(\vec{x}) = A\vec{x}$ is called a shear transformation. It can be shown that if T acts on each point in the 3×5 rectangle to the left, then the set of images forms the sheared parallelogram.

Set $k=1$:

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 5 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 8 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

The key idea is that the base of the rectangle stays, and every other point is sheared to one side.

The constant k , is called the shear factor.

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V

A transformation is linear if

$$(i) \quad T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v}) \text{ for all } \vec{u}, \vec{v} \text{ in the domain of } T \quad (1)$$

$$(ii) \quad T(c\vec{u}) = cT(\vec{u}) \text{ for all scalars } c \text{ and all } \vec{u} \text{ in the domain of } T \quad (2)$$

If T is a linear transformation, then

$$T(\vec{0}) = \vec{0} \quad (3)$$

and

$$T(c\vec{u} + d\vec{v}) = cT(\vec{u}) + dT(\vec{v}) \quad (4)$$

for all vectors $\vec{u}, \vec{v}$ in the domain of T and all scalars c, d .

Observe that if a transformation satisfies (4) for all $\vec{u}, \vec{v}$ and c, d , it must be linear. (Set $c=d=1$ for preservation of addition, and set $d=0$ for preservation of scalar multiplication.)

Repeated application of (4) produces a useful generalization:

$$T(c_1\vec{v}_1 + \cdots + c_p\vec{v}_p) = c_1T(\vec{v}_1) + \cdots + c_pT(\vec{v}_p) \quad (5)$$

In engineering and physics, (5) is referred to as a superposition principle. Think of $\vec{v}_1, \dots, \vec{v}_p$ as signals that go into a system and $T(\vec{v}_1), \dots, T(\vec{v}_p)$ as the responses of that system to the signals.

Property (3) follows from condition (ii), because $T(\vec{0}) = T(0\vec{u}) = 0T(\vec{u}) = \vec{0}$. Property (4) requires both (i) and (ii):

$$T(c\vec{u} + d\vec{v}) = T(c\vec{u}) + T(d\vec{v}) = cT(\vec{u}) + dT(\vec{v})$$

1.8 Introduction to Linear Transformations (cont'd)

VI

Example 4

Given a scalar r , define $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T(\vec{x}) = r\vec{x}$. T is called a contraction when $0 \leq r \leq 1$ and a dilation when $r > 1$. Let $r=3$ and show that T is a linear transformation.

Solution: Let $\vec{u}, \vec{v}$ be in $\mathbb{R}^2$, and let c, d be scalars. Then

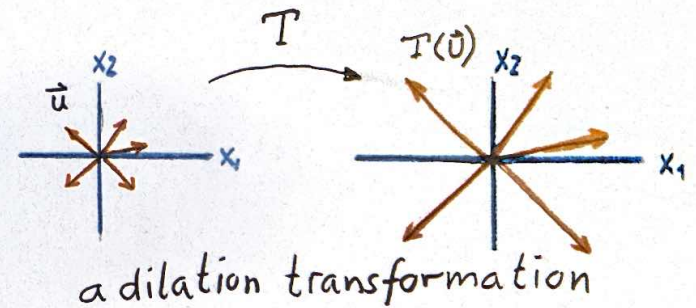
$$T(c\vec{u} + d\vec{v}) = 3(c\vec{u} + d\vec{v})$$

$$= 3c\vec{u} + 3d\vec{v}$$

$$= c(3\vec{u}) + d(3\vec{v})$$

$$= cT(\vec{u}) + dT(\vec{v})$$

Definition of T
Vector arithmetic



Example 5

Define a linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T(\vec{x}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -x_2 \\ x_1 \end{bmatrix}$.

Find the images under T of $\vec{u} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$, $\vec{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, and $\vec{u} + \vec{v} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$.

$$T(\vec{u}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}, \quad T(\vec{v}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \end{bmatrix}$$

$$T(\vec{u} + \vec{v}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 6 \\ 4 \end{bmatrix} = \begin{bmatrix} -4 \\ 6 \end{bmatrix}$$

It is obvious that $T(\vec{u} + \vec{v})$ is equal to $T(\vec{u}) + T(\vec{v})$.

T rotates the vectors ccw by 90° .

